

Assignment 1: AI System Specification

Course: Introduction to Artificial Intelligence

Assignment: PEAS Framework and Task Environment Analysis

Selected Application: Personalized News Feed Recommender (e.g., Twitter/X or TikTok Algorithm)

Name: Kapeel Dev

Roll No: 2K23/CSE/67

Task 1: PEAS Specification

1. Performance Measure

The success of a Personalized News Feed Recommender can be evaluated using the following metrics:

1. **User Engagement Rate** – Number of likes, comments, shares, and clicks generated by recommended content.
2. **Watch Time / Session Duration** – Total time users spend viewing recommended posts or videos.
3. **Recommendation Accuracy** – Degree to which recommended content matches user interests and preferences.
4. **User Retention Rate** – Percentage of users who continue using the platform over time.
5. **Content Diversity Score** – Ability to provide a variety of relevant content instead of repetitive recommendations.
6. **User Satisfaction** – Feedback measured through surveys, ratings, or reduced content-skipping behavior.

2. Environment

The recommender agent operates within a social media platform where users continuously interact with content.

Key External Factors:

- User interests and preferences
- Trending topics and viral content
- Content creators uploading new posts
- User demographic information
- Time of day and geographical location
- Platform policies and moderation rules

3. Actuators

The recommender system affects its environment through the following actions:

- Ranking posts in the user's feed
- Recommending videos, articles, or tweets
- Sending personalized notifications

- Highlighting trending content
- Adjusting content visibility and ordering
- Suggesting accounts or pages to follow

4. Sensors

The agent gathers information using:

- User click history
- Likes, shares, comments, and reactions
- Watch time and scrolling behavior
- Search history
- User profile information
- Device and location data
- Content metadata (hashtags, categories, and keywords)

Task 2: Environment Classification

Dimension	Classification	Justification
Fully Observable vs. Partially Observable	Partially Observable	The system cannot directly observe a user's true interests or intentions and must infer them from behavior.
Single-Agent vs. Multi-Agent	Multi-Agent	The recommender interacts with millions of users and content creators whose actions influence outcomes.
Deterministic vs. Stochastic	Stochastic	User responses to recommendations are uncertain and cannot be predicted with complete accuracy.
Episodic vs. Sequential	Sequential	Current recommendations influence future user behavior and recommendation decisions.
Static vs. Dynamic	Dynamic	User preferences, trends, and available content continuously change over time.
Discrete vs. Continuous	Continuous	User interactions and engagement metrics occur continuously and change in real time.

Task 3: Critical Analysis

1. Which Dimension Poses the Greatest Technical Challenge and Why?

The **Partially Observable** nature of the environment presents the greatest challenge for AI engineers. Users do not explicitly reveal all their interests, preferences, or future intentions. Instead, the recommender system must infer these hidden preferences from indirect signals such as clicks, watch time, likes, shares, and comments.

These behavioral signals can often be misleading because users may interact with content out of curiosity rather than genuine interest. Consequently, developing an accurate model of user preferences becomes a complex machine learning problem that requires continuous learning, adaptation, and refinement.

2. Utility Function and Trade-off Analysis

A suitable utility function for a personalized news feed recommender is:

$$\text{Utility} = (0.7 \times \text{User Engagement}) + (0.3 \times \text{Content Diversity})$$

This utility function balances the trade-off between maximizing engagement and ensuring users receive diverse recommendations.

If the weight assigned to **Content Diversity** is doubled, the utility function becomes:

$$\text{Utility} = (0.7 \times \text{User Engagement}) + (0.6 \times \text{Content Diversity})$$

Under this revised utility function, the recommender system would place greater emphasis on presenting a wider variety of content rather than focusing solely on highly engaging posts. Users would be exposed to different topics, viewpoints, and creators, thereby reducing the likelihood of creating an echo chamber.

Although overall engagement may decrease slightly in the short term, users are likely to benefit from a broader and more balanced content experience, potentially improving long-term satisfaction and retention.